Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 1 / 方法 1 [CO04-M01]

A construction may count witnesses rather than objects. Define the witness, compute its multiplicity, and inspect intersections before applying inclusion-exclusion. For extrema, an incidence bound is only half a solution: exhibit an arrangement satisfying every original condition.

一种构造可能数到的是见证而非对象。须定义见证、计算其重数，并在容斥前检查交集。对最值，关联计数界只是解答的一半，还须构造满足原全部条件的配置。

Use exact event intersections and the complement

精确计算事件交集与补集

Procedure

1. Choose an ambient set of equally counted objects and define the forbidden or required events.

2. Describe every intersection in structural terms, such as a common divisor, fixed cells, or a restricted image set.

3. Add single events, subtract pairs, add triples, and continue; include the empty or full object where appropriate.

4. For an exactly-one condition, account for intersection multiplicity rather than using a union count unchanged.

通用步骤

1. 选择统一计数的全集，并定义禁用或要求的事件。

2. 用结构描述每个交集，例如公约数、固定格或限制像集。

3. 单事件相加、两两相减、三者相加，依次进行，并妥善处理空对象或全对象。

4. 恰一事件发生时须调整交集重数，不能直接使用并集数。

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 1 / 方法 1 [CO04-M01] (continued / 续)

> Example. Among integers one through sixty, how many are divisible by exactly one of four and nine?

> 例题：一至六十的整数中，有多少个恰能被四、九之一整除?

Worked solution

1. There are $\lfloor 60/4 \rfloor = 15$ multiples of four and $\lfloor 60/9 \rfloor = 6$ multiples of nine.

2. The intersection is the multiples of $\operatorname{lcm}(4, 9) = 36$, of which there is one.

3. For exactly one event, subtract that intersection twice: $15 + 6 - 2 = 19$.

例题解析

1. 四的倍数有 $\lfloor 60/4 \rfloor = 15$ 个，九的倍数有六个。

2. 交集是最小公倍数三十六的倍数，只有一个。

3. 恰一事件须减去交集两次，得到 $15 + 6 - 2 = 19$。

Intersections of divisor sets are divisors of a gcd; intersections of multiple sets are multiples of an lcm.

约数集合的交集是最大公因数的约数集，倍数集合的交集是最小公倍数的倍数集。

Practice / 配套练习: AIME-2001-II-Q09 (WB p. 735), AIME-2002-I-Q01 (WB p. 721), AIME-2002-II-Q09 (WB p. 633), AIME-2005-II-Q04 (WB p. 602), AIME-2006-II-Q07 (WB p. 586), AIME-2007-II-Q10 (WB p. 738), AIME-2008-II-Q12 (WB p. 654), AIME-2009-II-Q09 (WB p. 547), AIME-2013-II-Q09 (WB p. 672), AIME-2014-I-Q11 (WB p. 792), AIME-2014-I-Q12 (WB p. 746), AIME-2015-I-Q09 (WB p. 150), AIME-2016-II-Q11 (WB p. 611), AIME-2017-II-Q01 (WB p. 660), AIME-2017-II-Q11 (WB p. 702), AIME-2021-II-Q06 (WB p. 640), AIME-2022-I-Q13 (WB p. 598), AIME-2024-II-Q11 (WB p. 61), AIME-2025-I-Q04 (WB p. 539), AIME-2025-II-Q07 (WB p. 492)

Section 4.4 Inclusion-exclusion & double counting

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Method 2 / 方法 2 [CO04-M02]

Classify compatible overlapping patterns

分类相容的重叠模式

Procedure

1. Specify each local bad pattern, including its direction and any surrounding separator.

2. For selected patterns, prove whether they can coexist and how many positions or labels their union fixes.

3. Glue overlapping adjacency runs or count the free remaining cells, then apply inclusion-exclusion.

4. Handle patterns that wrap around a circle and the all-selected exceptional object separately.

通用步骤

1. 明确每个局部坏模式，包括方向与周围分隔。

2. 对选中的模式，证明能否共存以及并集固定多少位置或标签。

3. 把重叠相邻关系粘成连续段，或计算剩余自由格，再容斥。

4. 单独处理绕圆闭合的模式与全部选中的例外对象。

> Example. How many permutations of A, B, C, D contain neither directed adjacent block AB nor directed adjacent block BC?

> 例题：字母 A, B, C, D 的排列中，有多少个既不含有向相邻块 AB，也不含 BC?

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 2 / 方法 2 [CO04-M02] (continued / 续)

Worked solution

1. The ambient count is $4! = 24$. Each forbidden block, glued as one unit, gives $3! = 6$ permutations.

2. Both blocks can coexist only as ABC, which is one block together with D, giving two permutations.

3. Inclusion-exclusion gives $24 - 6 - 6 + 2 = 14$. The overlap is a single ordered run, not two freely orientable blocks.

例题解析

1. 全集有 $4! = 24$ 个排列，每个禁块粘成一单位后有 $3! = 6$ 种。

2. 两块共存只能形成 ABC，与 D 共两个单位，给出两种排列。

3. 容斥得 $24 - 6 - 6 + 2 = 14$。重叠是一个定向连续段，不是两个可自由转向的区块。

Overlapping witnesses may force an extra occurrence; prove that your multiplicity correction covers it.

重叠见证可能迫使额外出现，须证明重数修正覆盖全部情况。

If selected undirected forbidden edges on a line form r nonempty runs and contain k edges, they collapse to $N - k$ blocks and contribute $2^r(N - k)!$. A circular long run can be marked by its preceding unselected position; this excludes the all-selected object, which must be restored separately.

For fixed-sum pair constraints on a small selected set, model each pair as an edge. Shared endpoints form path witnesses, while disjoint pairs form matching witnesses. List possible witness multiplicities before adding their counts; a set with two witnesses must lose one copy, not two.

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 2 / 方法 2 [CO04-M02] (continued / 续)

直线上选中的无向禁邻边若共 k 条且形成 r 个非空连续段，会粘成 $N-k$ 个区块，贡献 $2^r(N-k)!$。圆上的长段可由它前一个未选位置标记；这会排除全选对象，须另外补回。

对固定和二元组限制，把二元组视为边。共端点形成路径见证，不交边形成匹配见证。相加前列出可能见证重数；有两个见证的对象应扣去一份重复，而非扣去两份。

Practice / 配套练习: AIME-2001-II-Q09 (WB p. 735), AIME-2010-II-Q11 (WB p. 667), AIME-2011-II-Q12 (WB p. 745), AIME-2014-II-Q09 (WB p. 663), AIME-2018-I-Q09 (WB p. 664), AIME-2020-II-Q09 (WB p. 665), AIME-2022-I-Q06 (WB p. 666)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 3 / 方法 3 [CO04-M03]

Mark an incidence to sum weights or powers

标记关联以求权重或幂次总和

Procedure

1. Replace a weight by the number of marks that produce it.

2. Count pairs or tuples consisting of an object and its marks in the more convenient order.

3. Separate coincident marks from distinct marks when taking a square or higher moment.

4. For geometric incidences, count underlying pairs rather than assuming one pair per visible point.

通用步骤

1. 把权重替换为产生该权重的标记数。

2. 按更方便的顺序计算对象与标记组成的二元或多元组。

3. 平方或更高次矩须区分标记相同与不同。

4. 几何关联应计底层对象对，不能默认每个可见交点只对应一对。

> Example. Compute $\sum_{A\subseteq\{1,\ldots,5\}} |A|^2$ without listing subsets.

> 例题：不逐个列子集，计算 $\sum_{A\subseteq\{1,\ldots,5\}} |A|^2$。

Worked solution

1. For each subset, $|A|^2$ counts ordered marked pairs $(u, v) \in A^2$.

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 3 / 方法 3 [CO04-M03] (continued / 续)

2. When $u = v$, choose the mark in five ways and the other four elements freely, giving $5 \cdot 2^4 = 80$.

3. When $u \neq v$, choose $5 \cdot 4$ ordered marks and the other three elements freely, giving160. Total 240.

例题解析

1. 对每个子集，$|A|^2$ 计算有序标记对 $(u, v) \in A^2$。

2. 若两标记相同，先选五种标记，其余四元素自由选，贡献八十。

3. 若不同，选 $5 \cdot 4$ 个有序标记，其余三元素自由选，贡献一百六十，总和二百四十。

A point where k lines concur consumes $\binom{k}{2}$ line pairs.

有 k 条直线共点时，该点占用 $\binom{k}{2}$ 个直线对。

For an incidence-weighted subset sum, fix one marked element first and count the remaining choices. For a sum of valuations in a product, count one contribution from each multiple of p, another from each multiple of p^2, and so on. These are the same object-mark counting principle.

子集的关联加权和可先固定一个标记元素，再计其余选择。乘积的赋值总和则可对每个 p 的倍数计一次、每个 p^2 的倍数再计一次，依此类推；本质都是对象与标记的双计数。

Practice / 配套练习: AIME-2003-I-Q03 (WB p. 658), AIME-2010-II-Q03 (WB p. 468), AIME-2022-I-Q12 (WB p. 668), AIME-2023-I-Q03 (WB p. 661), AIME-2024-II-Q01 (WB p. 662)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 4 / 方法 4 [CO04-M04]

Correct nonuniform witness multiplicities

修正不均匀的见证重数

Procedure

1. Determine how many witnesses each finished object has.

2. Partition into generic and exceptional multiplicity classes.

3. Convert incidence counts to object counts using those exact multiplicities, verifying that all exceptions have been listed.

通用步骤

1. 确定每个完成对象有几个见证。

2. 按一般重数与例外重数分类。

3. 用精确重数把关联数转为对象数，并证明例外已经列全。

> Example. Choose three elements from $\{1,2,3,4,5,6\}$ and require at least one even element. A construction first marks an even selected element. Why can its count not simply be used as the answer?

> 例题：从一至六中选三数，要求至少一个偶数。若构造先标记一个被选偶数，为什么不能直接把构造数当答案?

Worked solution

1. With exactly $r = 1, 2, 3$ even elements, the subset counts are $\binom{3}{r}\binom{3}{3-r} = 9, 9, 1$.

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 4 / 方法 4 [CO04-M04] (continued / 续)

2. Marking an even element counts each such subset r times, so the incidence total is $9 + 18 + 3 = 30$.

3. The desired object count is $9 + 9 + 1 = 19$, also $\binom{6}{3} - 1$. The overcount is nonuniform.

例题解析

1. 恰有 $r = 1, 2, 3$ 个偶数时，子集数为 $\binom{3}{r}\binom{3}{3-r} = 9, 9, 1$。

2. 标记一个偶数使每个子集被数 r 次，关联总数为 $9 + 18 + 3 = 30$。

3. 真正对象数为 $9 + 9 + 1 = 19$，也即 $\binom{6}{3} - 1$；重复倍数并不统一。

Three equal sides make an equilateral triangle have three possible apex witnesses; a generic isosceles triangle has one.

等边三角形有三个顶角见证，一般等腰三角形只有一个。

Practice / 配套练习: AIME-2002-I-Q05 (WB p. 690), AIME-2004-I-Q03 (WB p. 691), AIME-2017-II-Q13 (WB p. 703), AIME-2017-II-Q14 (WB p. 704), AIME-2020-I-Q05 (WB p. 622), AIME-2024-II-Q11 (WB p. 61)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 5 / 方法 5 [CO04-M05]

Double count a capacity bound and construct equality

双计数容量界并构造等号

Procedure

1. Count the same total across two families of layers, vertices, or groups.

2. Combine the resulting identities with integrality or a capacity bound.

3. Determine the equality requirements, then build one configuration satisfying every layer or group condition.

通用步骤

1. 从两族层、顶点或群体分别计算同一总量。

2. 结合所得恒等式与整数性或容量界。

3. 找出等号要求，再构造同时满足每层每组条件的配置。

> Example. A zero-one matrix has three rows and five columns. Each row has two ones, and each column has at most two. What is the least possible number of nonzero columns?

> 例题：三行五列零一矩阵每行有两个一，每列至多两个一，非零列最少有多少列？

Worked solution

1. Row counting gives six ones in total. With at most two per nonzero column, at least three columns are needed.

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 5 / 方法 5 [CO04-M05] (continued / 续)

2. Use rows $11000, 10100, 01100$. Each row has two ones and the first three columns each have two.

3. Thus three nonzero columns attain the lower bound; the unused columns do not violate any condition.

例题解析

1. 按行计共有六个一，非零列各至多两个，所以至少三列。

2. 取三行为 $11000, 10100, 01100$，各行两个一，前三列各两个一。

3. 故三列达到下界，其余空列不违反条件。

An integer dimension satisfying the counting equations may still need an explicit color or incidence construction.

满足计数方程的整数尺寸仍可能需要显式颜色或关联构造。

Practice / 配套练习: AIME-2001-II-Q02 (WB p. 657), AIME-2016-II-Q04 (WB p. 659)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 6 / 方法 6 [CO04-M06]

Count cyclic overlaps before analyzing exact exchange

先计循环重叠再分析完全交换

Procedure

1. For a subset B of a cyclic group, count ordered pairs of distinct selected points by their displacement.

2. Use $\sum_{k\neq 0} |B \cap (B+k)| = |B|(|B|-1)$ to force a zero overlap when the average is below one.

3. At half the positions, zero overlap means the shift exchanges the set with its complement; analyze each rotation cycle and its parity.

通用步骤

1. 对循环群子集 B，按位移计算不同所选点的有序对。

2. 利用 $\sum_{k\neq 0} |B \cap (B+k)| = |B|(|B|-1)$，平均值小于一时必有零重叠。

3. 若恰选一半位置，零重叠意味着平移交换集合与补集；再分析各旋转轨道及其奇偶性。

> Example. Three positions are selected on a labeled ten-cycle. Prove that at least three nonidentity rotations make the selected set disjoint from its image, and show this bound can be sharp.

> 例题：十个有标识循环位置中选三个，证明至少三个非恒等旋转使所选集与其像不交，并说明此界可达到。

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 6 / 方法 6 [CO04-M06] (continued / 续)

Worked solution

1. The nine nonzero displacement overlaps total $3 \cdot 2 = 6$. Therefore at most six rotations have a positive overlap, leaving at least three zero-overlap rotations.

2. For $B = \{0, 1, 3\}$ modulo ten, its six directed nonzero differences are $1, 2, 3, 7, 8, 9$, all distinct.

3. Exactly rotations by $4, 5, 6$ have zero overlap, so the lower bound three is sharp.

例题解析

1. 九个非零位移的重叠总数为 $3 \cdot 2 = 6$，至多六个旋转有正重叠，所以至少三个无重叠。

2. 取模十子集 $B = \{0, 1, 3\}$，六个非零有向差为 $1, 2, 3, 7, 8, 9$，彼此不同。

3. 恰有位移四、五、六无重叠，因此三个的下界可达到。

Complement exchange requires alternating membership on every cycle, so every cycle must have even length.

交换集合与补集要求每个轨道上成员交替出现，故每个轨道长度必须为偶数。

For a half-size subset, a shift k partitions the cycle into $\gcd(n, k)$ cycles of length $n/\gcd(n, k)$. If that length is even, complement-exchanging colorings number $2^{\gcd(n,k)}$. These families for different shifts may overlap, so do not add them without checking intersections.

选半数位置时，位移 k 将循环分成 $\gcd(n, k)$ 个长度 $n/\gcd(n, k)$ 的轨道。长度为偶数时，交换补集的染色有 $2^{\gcd(n,k)}$ 种。但不同位移对应的染色族可能重叠，不能未经交集检验直接相加。

Practice / 配套练习: AIME-2024-I-Q11 (WB p. 669)